Hamed Hojatian

Ph.D. Candidate

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Education

Fall 2019 – Now
Polytechnique Montréal
Ph.D. in AI-aided wireless communication systems
Supervisors: Prof. François Leduc-Primeau, Prof. Jean-François Frigon
GPA: 4.0/4.0

2014-2017
Isfahan University of Technology (IUT)
M.Sc. in communication system engineering
Supervisor: Prof. Mohammad Javad Omidi
GPA: 3.7/4.0

2010-2014
Babol (Noshirvani) University of technology (NIT)
B.Sc. in electrical engineering
Supervisor: Prof. Hossein Miar-Naeimi
GPA: 3.1/4.0

Research Experience

Ph.D.
AI-aided Beamforming in massive MIMO Systems
- RSSI-Based hybrid beamforming design with deep learning;
- Improve the spectral and energy efficiency of hybrid beamforming design;
- Unsupervised learning for hybrid beamforming design and intelligent beamforming;
- Coordinated hybrid beamforming and cell-free massive MIMO with deep neural network; and
- Subarray hybrid beamforming and efficiecnt hardware design driven by unsupervised learning.
- ...

M.Sc.
Time and Frequency synchronization in OFDM-based Massive MIMO systems
- Time and frequency synchronization in OFDM-based massive MIMO;
- Carrier Frequency Offset (CFO) and channel estimation;
- Detection in receiver using iterative method; and
- Improving the performance by using Filtered-OFDM.

B.Sc.
OFDM Transmitter and receiver simulation
- Developed in MATLAB simulation
- OFDMA multi-user simulation in AWGN channel.

Skills

Tools and Languages	Python, Pytorch, TensorFlow, MATLAB, HTML and CSS, JavaScript, Bash, and Microsoft office.
Telecomm. Tools	Ericsson OSS, MapInfo, Atoll, TEMS, Actix
Languages	English, French (basic), Persian (native)

Community Service

IEEE Journals reviewer	Selected Areas in Communications (**J-SAC**), Transactions on Wireless Communications (**TWC**), Transactions on Communications (**TCOMM**), Wireless Communications Letters (**WCL**), Communications Letters (**CommL**), and Transactions on Vehicular Technology (**TVT**). Details in here.

Research Interests

- AI-aided wireless communication;
- Machine Learning, Deep Learning, DRL, Federated Learning, META and few shot learning;
- Statistical learning, optimization;
- Signal processing in massive MIMO communication, 5G beyond and 6G;
- Intelligent beamforming, coordinated beamforming, distributed antennas, cell free communication; and
- Millimeter wave in Massive MIMO and antenna array.

Work Experience

Nov 2022 - present

Ericsson
Research intern, Co-op research

Nov 2015 - Nov 2018

Mobile Telecommunication Company of Iran (MCI)
Radio planning and optimization engineer

- Live network Ericsson OSS optimization and parameter tuning for 2G, 3G, and LTE;
- Network Planning (2G, 3G and LTE), new site baseline parameters tuning and planning;
- Project manager of RNC rehoming and RNC planning; and
- Project manager of site renaming.

Oct 2015

Eleventh (11th) national robotics competition at BABOL University of Technology
Vice-President

Publications

2022

1. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Decentralized Beamforming for Cell-Free Massive MIMO with Unsupervised Learning**". ***IEEE Communications Letters*** (2022).
2. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Flexible Unsupervised Learning for Massive MIMO Subarray Hybrid Beamforming**". ***IEEE GLOBECOM*** (2022).

2021

3. **Hojatian, H.**, Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**Unsupervised Deep Learning for Massive MIMO Hybrid Beamforming**". ***IEEE Transactions on Wireless Communications*** (2021).

2020

4. **Hojatian, H.**, Nguyen Ha, V., Nadal, J., Frigon, J.-F. & Leduc-Primeau, F. "**RSSI-Based Hybrid Beamforming Design with Deep Learning**". ***IEEE ICC*** (2020).

2016

5. **Hojatian, H.**, Omidi, M. J., Saeedi-Sourck, H. & Farhang, A. "**Joint CFO and Channel Estimation in OFDM-based Massive MIMO Systems**". ***IEEE IST*** (2016).

Talks and Teaching Experience

2022 Hybrid-Seminars on Selected Topics in Signal Processing.
Presenting my research progress in **IVADO - Digital October.**

2021 Presenting my research progress in **IVADO - Digital October.**

2020 Paper Presentation, **IEEE ICC 2020**, Dublin, Ireland.
Presenting my research topic in **IVADO - Digital October.**

2013 Teaching assistant for **Electronic I**.
Teaching assistant for **Electronic circuit I**.

2012 Teaching assistant for **Digital Signal Processing**.

Certificates

2022 Introduction to Unix Shell, **Calcul Quebec**.
Massive data analysis with Spark, **Calcul Quebec**.
Introduction to GPU Programming with Python, **Calcul Quebec**.

2019 4th **IVADO/MILA** SCHOOL IN DEEP LEARNING.

2018 DEEP LEARNING WITH PYTORCH, **Udemy**.
LTE OPTIMIZATION OF RADIO NETWROK (3GPP), **TechCom**.

2017 LTE COUNTER AND KPI (ERICSSON), **TechCom**.
LTE RADIO NETWORK PLANNING, **TechCom**.

Mentoring

2019 Master student at Polytechnique Montréal.
Internship research student from Cameron.

Strengths

Hard-working, Decision Making, Ability to Work Under Pressure, Motivator & Leader Conflict Resolution, Time Management, Target oriented, Flexibility, and Adaptability